

Wi-Fi Series

Data Sheet

USE_8189

Wi-Fi Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	19/05/2026	MB		Initial release

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Features :

General

- Support Wi-Fi functionalities
- IEEE 802.11b/g/n compliant
- Single-band 1T1R architecture
- SDIO/GPIO interface for WLAN
- Dimension: 12 × 12 × 1.8 mm
- Operating temperature: 0°C to 70°C
- Storage temperature: -40°C to 80°C

MCU Platform

- Chipset:RTL8189FTV-VQ1

WLAN Features

- IEEE standards support: IEEE 802.11b, IEEE 802.11g, IEEE 802.11n
- WiFi 1T1R supports data rates up to 72.2 Mbps
- Operate at ISM frequency bands (2.4GHz)

Host Interface

- SDIO V1.1/2.0 Interface for Wi-Fi

General Description:

The USE_8189 is a highly integrated and excellent performance Wireless LAN (WLAN) SDIO network interface device.

This WLAN module design is based on Realtek RTL8189FTV-VQ1. It is a highly integrated single-chip 1T1R SISO Wireless LAN (WLAN) SDIO network interface controller complying with the IEEE 802.11b/g/n specification.

It combines MAC, baseband, and RF in a single chip. The module is designed to provide excellent wireless performance with low power consumption and cost-effective implementation.

The wireless module supports high-speed wireless connection up to 72.2 Mbps.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

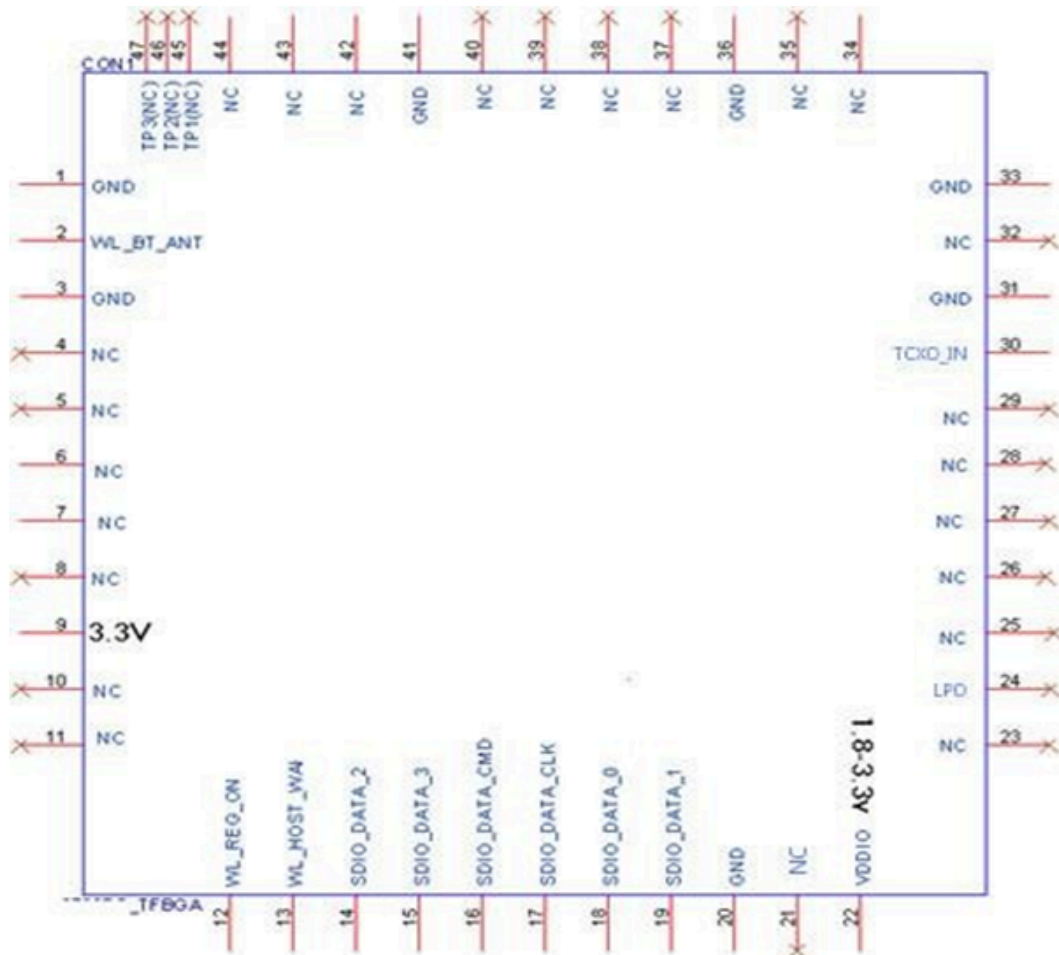


Fig1: Module Pinouts

Pin Descriptions:

Pin No	Pin Name	Type	Description
1	GND	—	Ground connections
2	RF	I/O	RF OUTPUT
3	GND	—	Ground connections
4~8	NC	—	Floating (NC)
9	3.3V	P	Main power input
10	NC	—	Floating (NC)
11	NC	—	Floating (NC)
12	WL_REG_ON	I	Chip_EN, external pull low shutdown RTL8189FTV
13	WL_HOST_WAKE	O	WLAN WAKE HOST
14	SDIO_DATA_2	I/O	SDIO_D2
15	SDIO_DATA_3	I/O	SDIO_D3
16	SDIO_DATA_CMD	I/O	SDIO_CMD
17	SDIO_DATA_CLK	I	SDIO_CLK
18	SDIO_DATA_D0	I/O	SDIO_D0
19	SDIO_DATA_D1	I/O	SDIO_D1
20	GND	—	Ground connections
21	NC	—	Floating (NC)
22	VDIO	P	1.8V or 3.3V
23	NC	—	Floating (NC)
24	LPO	—	CLK_REQ, not used please NC
25	NC	—	Floating (NC)
26~29	NC	—	Floating (NC)

30	TCXO_IN	—	26MHz_IN, not used please NC
31	GND	—	Ground connections
32	NC	—	Floating (NC)
33	GND	—	Ground connections
34~35	NC	—	Floating (NC)
36	GND	—	Ground connections
37~40	NC	—	Floating (NC)
41	GND	—	Ground connections
42~44	NC	—	Floating (NC)

The signal type codes are used in the table:

- P: POWER
- I: INPUT
- O: OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	0	25	70	deg.C
VCC33	3.0	3.3	3.6	V
VDDIO	1.62	1.8 or 3.3	3.6	V

Power Consumption

Mode	Status	Power(mA)	Note
OS Windows XP	Link	130	
	RX	130	20M
	TX	190	20M(MCS7)

RF Characteristics

Wi-Fi Specification

Feature	Description	
WLAN Standard	IEEE 802.11 b/g/n Wi-Fi compliant	
Frequency Range	2.4 GHz ~ 2.4835GHz (2.4 GHz ISM Band)	
Number of Channels	2.4GHz : Ch1 ~ Ch14	
Test Items	Typical Value	EVM
Output Power	802.11b /11Mbps : 16dBm $\pm$ 2 dB	EVM $\leq$ -10dB
	802.11g /54Mbps : 14dBm $\pm$ 2 dB	EVM $\leq$ -25dB
	802.11n /MCS7 : 13dBm $\pm$ 2 dB	EVM $\leq$ -28dB
Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	$\pm$ 20ppm	
SISO Receive Sensitivity (11b,20MHz) @8% PER	- 11Mbps PER @ -84 dBm	$\leq$ -76
SISO Receive Sensitivity (11g,20MHz) @10% PER	- 54Mbps PER @ -72 dBm	$\leq$ -68
SISO Receive Sensitivity (11n,20MHz) @10% PER	- MCS=7 PER @ -67 dBm	$\leq$ -65

SDIO Interface Timing :

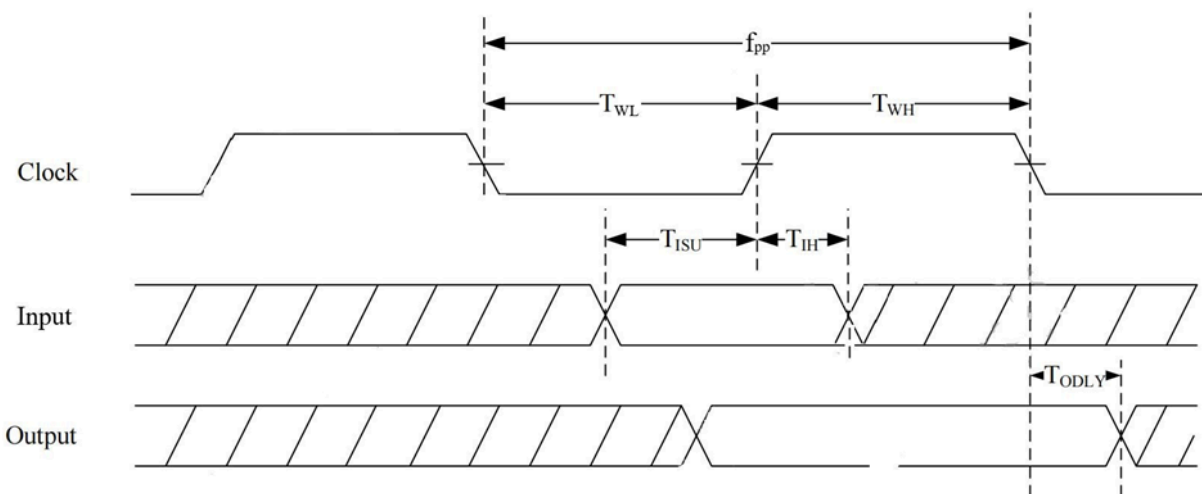


Fig 2: SDIO/GSPI Interface Timing

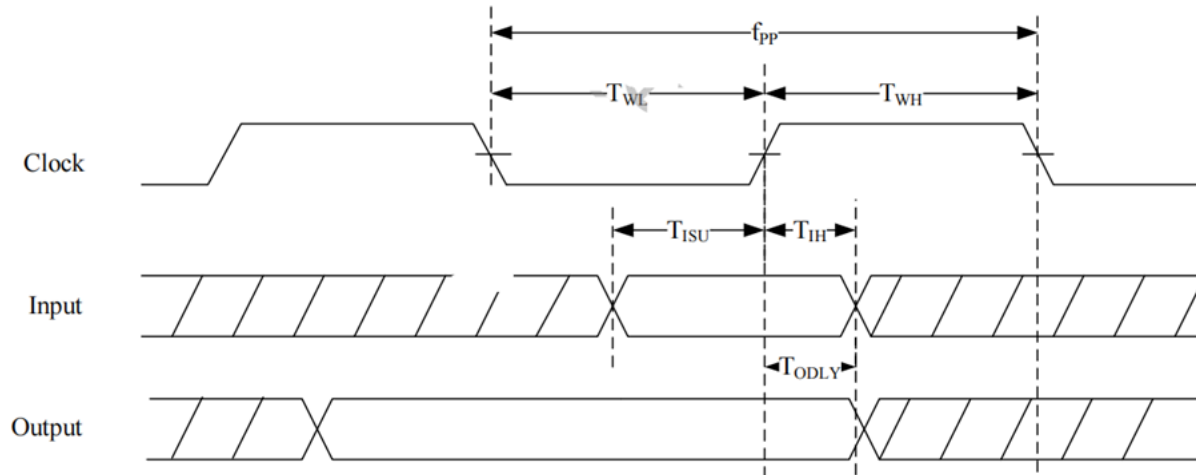


Fig 3: SDIO High Speed Interface Timing

SDIO/GSPI Interface Timing Parameters

NO	Parameter	Mode	MIN	MAX	Unit
f_{pp}	Clock Frequency	Default	0	25	MHz
		HS	0	50	MHz
T_{WL}	Clock Low Time	DEF	10	-	ns
		HD	7	-	ns
T_{WH}	Clock High Time	DEF	10	-	ns
		HD	7	-	ns
T_{ISU}	Input Setup Time	DEF	5	-	ns
		HD	6	-	ns
T_{IH}	Input Hold Time	DEF	5	-	ns
		HD	2	-	ns
T_{ODLY}	Output Delay Time	DEF	-	14	ns
		HD	-	14	ns

SDIO Interface Power-On Sequence

After power-on, the SDIO interface is selected by the RTL8189FTV-VC-CG automatically when a valid SDIO command is received. To attain better SDIO host compatibility, the following power-on sequence is recommended:

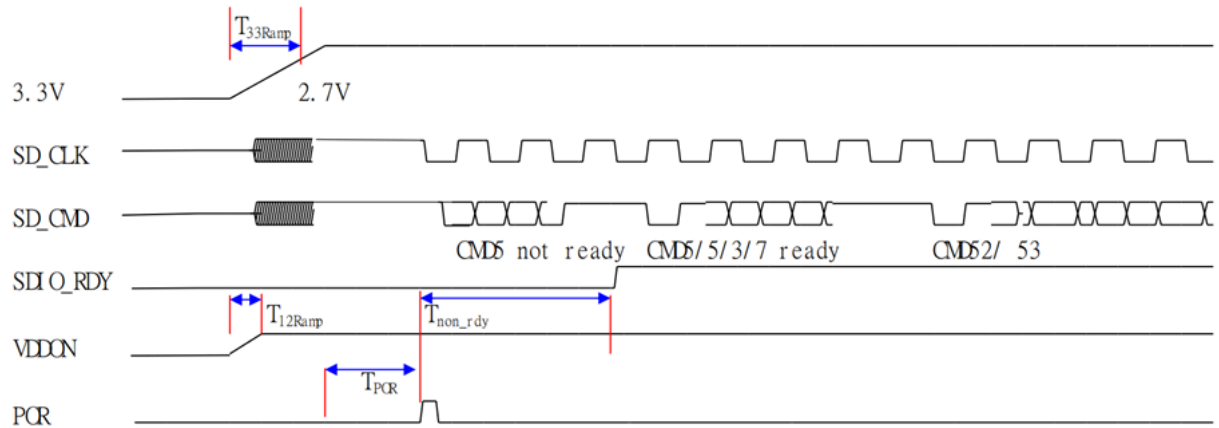


Fig4: SDIO Interface Power-On Sequence

Variable definition:

T_{33ramp} : The 3.3V main power ramp up duration.

T_{12ramp} : The internal 1.2V ramp up duration.

T_{POR} : The duration the power-on reset releases, and the power management unit executes power-on tasks. The power-on reset will detect both 3.3V and 1.2V power ramp up after a predetermined duration.

T_{non_rdy} : SDIO not ready duration. In this state the RTL8189FTV-VC-CG may respond to commands without the ready bit set. After the ready bit is set, the host will initiate the full card detection procedure

Power-On Flow Description

The 3.3V ramp up time is specified by the T_{33ramp} duration.

After 3.3V ramp up and 1.2V ramp up, the power management unit will be enabled by the power ready detection circuit, and will enable the SDIO block. eFUSE is then autoloading to the SDIO circuits during the T_{non_rdy} duration. After the autoloading has completed, the SDIO sets the ready bit. After CMD5/5/3/7 procedures, card detection is then executed. After the driver has loaded, normal commands 52 and 53 are then used.

A typical timing specification is shown below:

SDIO Interface Power-On Timing Parameters

Parameter	Min	Typical	Max	Unit
T_{33ramp}	0.25	0.5	2.5	ms
T_{12ramp}	0.1	0.5	1.5	ms
T_{por}	2	2	8	ms
T_{non_rdy}	1	2	10	ms

Power off by chip_en sequence

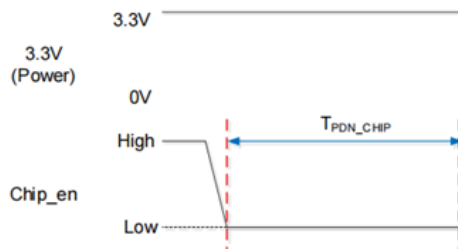


Figure 4. Power Off by chip_en Sequence

Table 12. Power Off by WL# and BT_DIS# Timing Parameters

	Min	Typical	Max	Unit	Description
T_{PDN_CHIP}	10	100	--	ms	Chip_en keep low duration

Chip_en can externally shutdown the RTL8189FTV –VQ1 when chip_en is pulled low. The keeping low duration must be more than T_{PDN_CHIP}

Power off by 3.3V power sequence

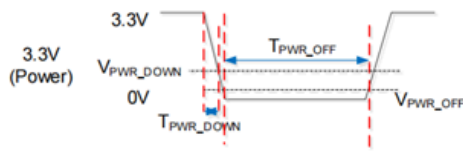


Figure 5. Power Off by 3.3V power Sequence

Table 13. Power Off by 3.3V power Timing Parameters

	Min	Typical	Max	Unit	Description
T_{PWR_OFF}	10	100	--	ms	3.3V power off time
V_{PWR_OFF}	--	--	0.4	V	3.3V power off voltage
V_{PWR_DOWN}	--	--	0.7	V	3.3V power down voltage
T_{PWR_DOWN}	--	--	10	ms	3.3V power down time

When 3.3V power off and on afterward, the voltage of 3.3V power must keep lower than V_{PWR_OFF} , and the 3.3V power keeping off duration must be more than T_{PWR_OFF}

SDIO Interface

Symbol	Type	Pin No	Description
SD_CLK	I	17	SDIO Clock Input
SD_CMD	I/O	16	SDIO Command Input
SD_D0	I/O	18	SDIO Data Line 0
SD_D1	I/O	19	SDIO Data Line 1

SD_D2	I/O	14	SDIO Data Line 2
SD_D3	I/O	15	SDIO Data Line 3

GPIO Interface

Symbol	Type	Pin No	Description
GSPI_CLK	I	17	GSPI Clock Input
GSPI_MOSI	I	16	GSPI Data Input
GSPI_MISO	O	18	GSPI Data Out
GSPI_SIRQ	O	19	GSPI Interrupt
GSPI_SCSn	I	15	GSPI Chip Select Bar

Block Diagram :

Single-Band 11n (1x1) Solution

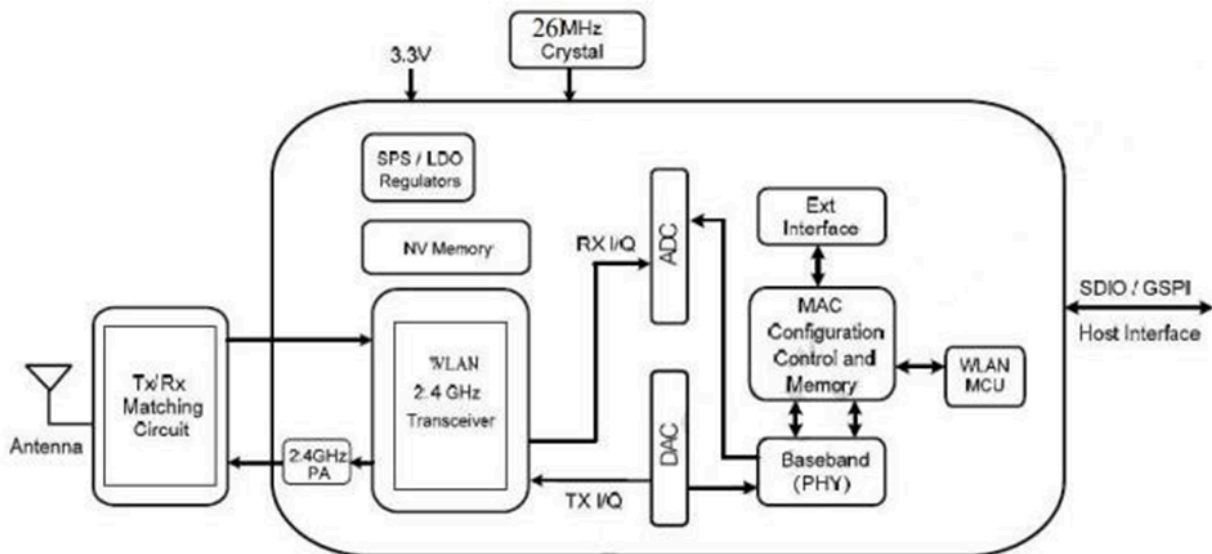


Fig6 : Block Diagram

Reference Design :

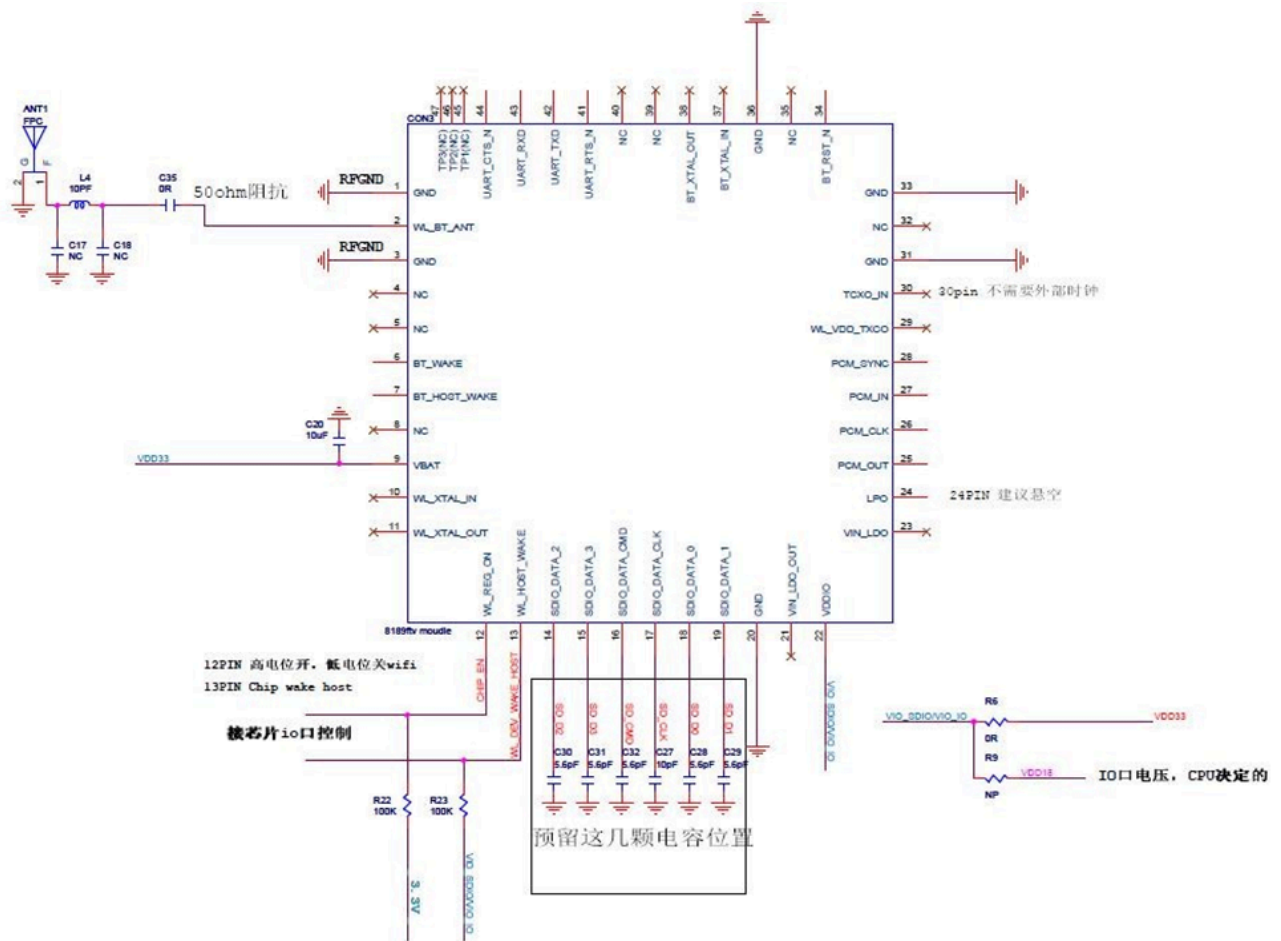


Fig7 : Reference Design

Physical Dimensions :

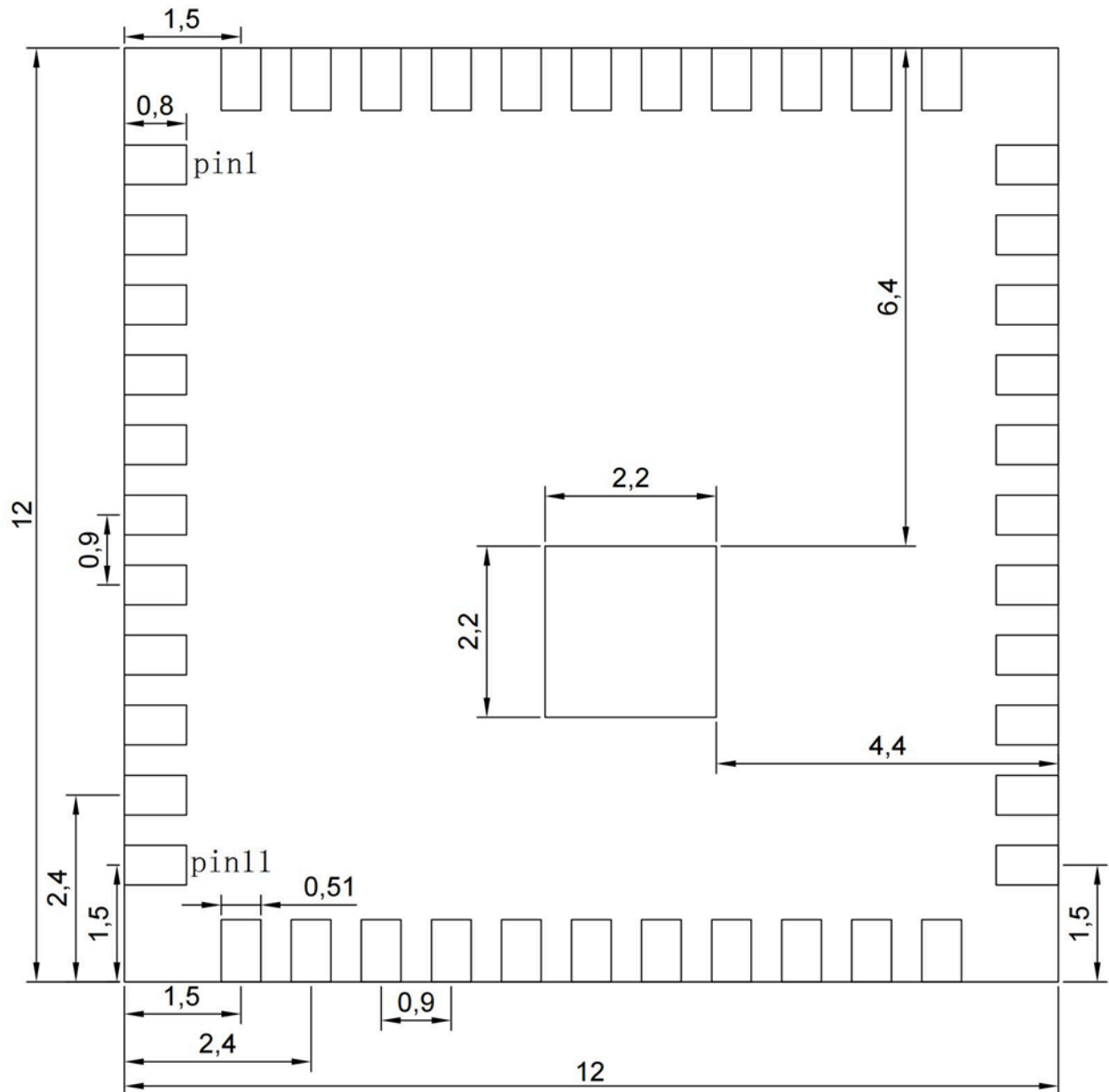


Fig8 : Physical Diagram (top view)

Layout Recommendation:

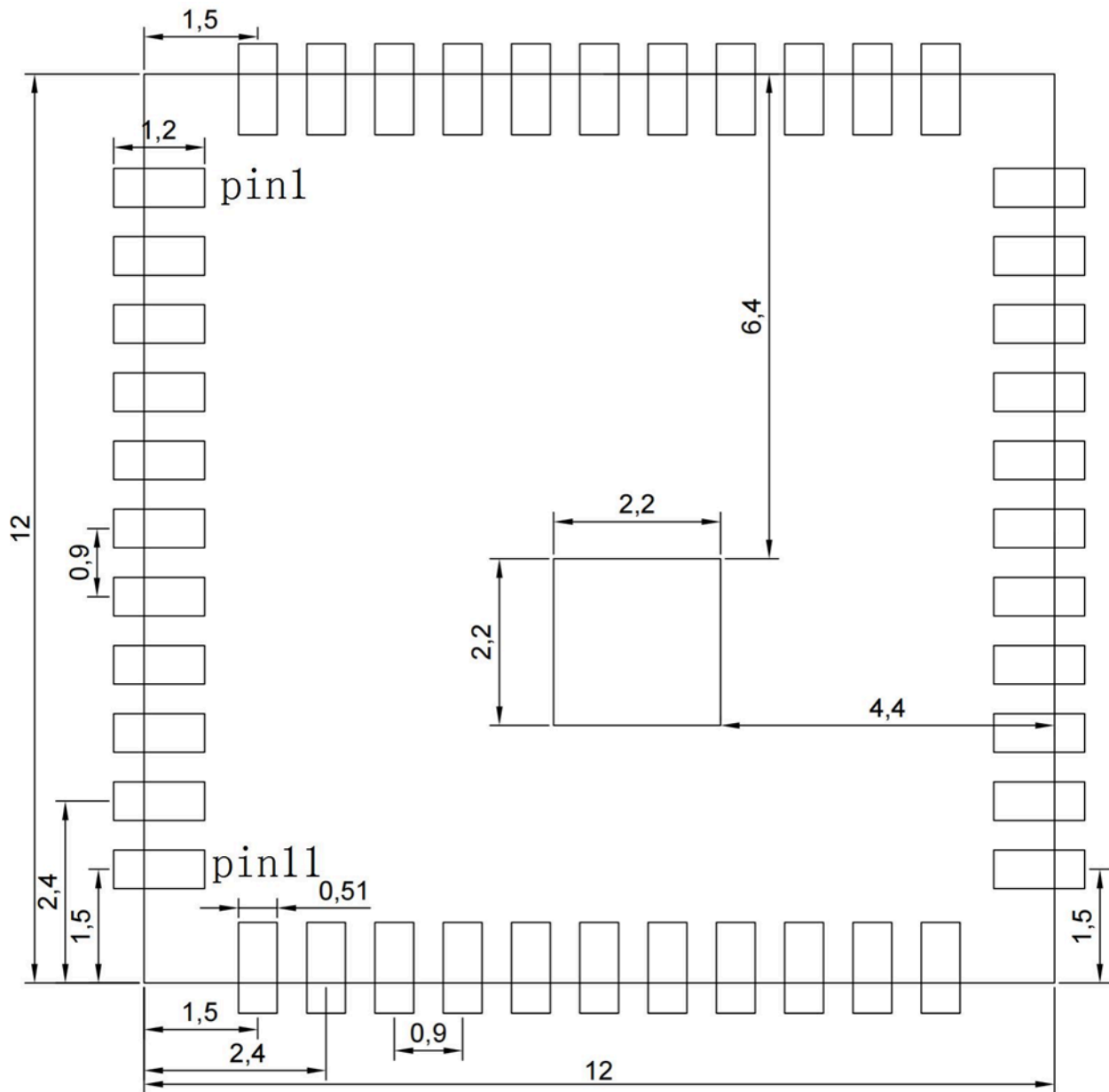


Fig9 : Layout Diagram

Module Dimension:

- L x W 12 x 12 (+0.3/-0.1) mm

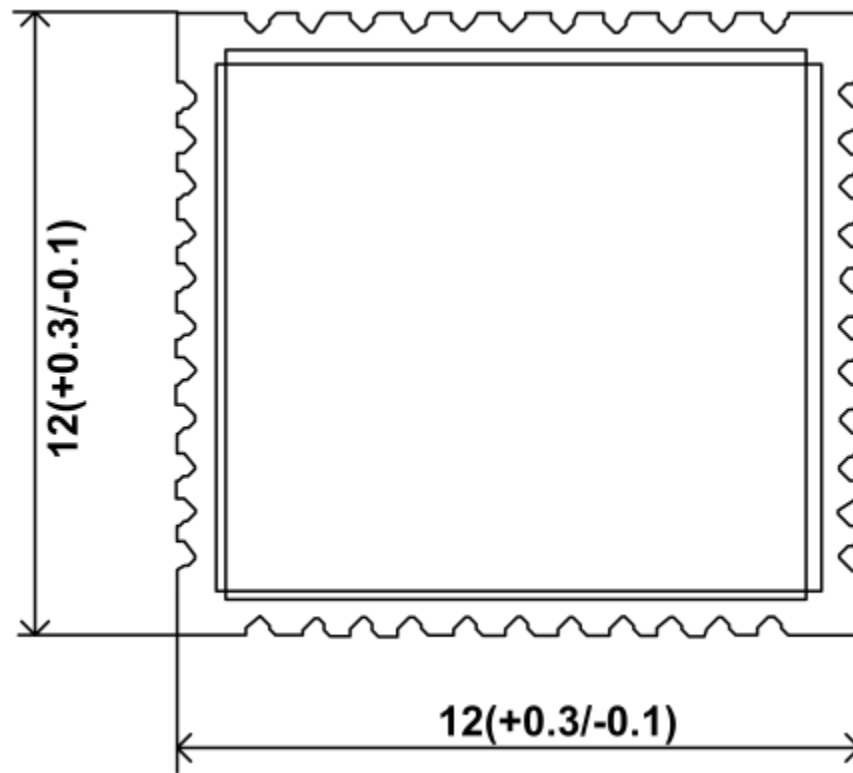


Fig10(a) : L x W

- H: 1.8 (±0.2) mm

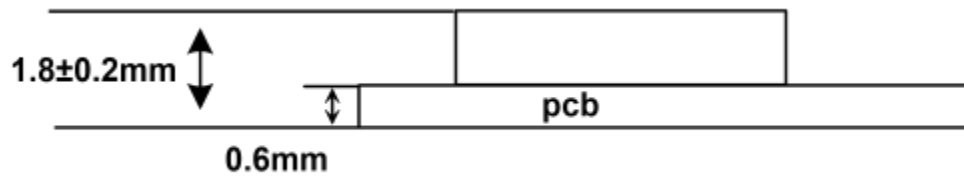


Fig10(b) : Height

Recommended Reflow Profile :

Referred to IPC/JEDECstandard.

Peak Temperature : <260°C Number

of Times : £2 times

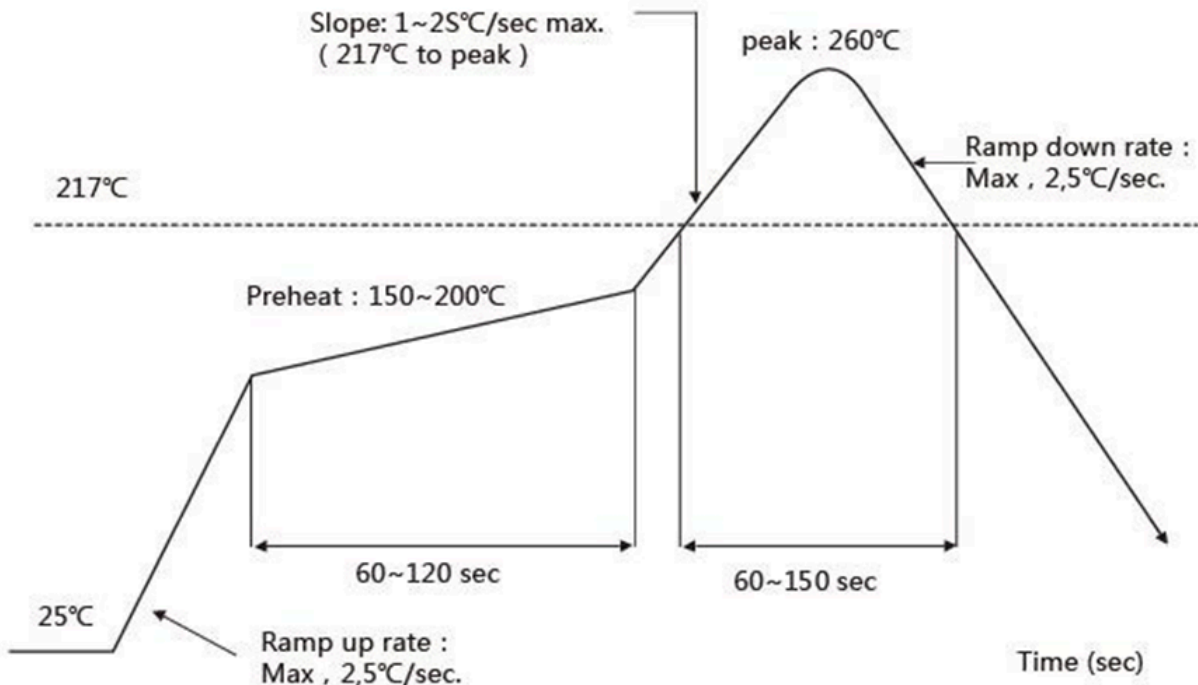
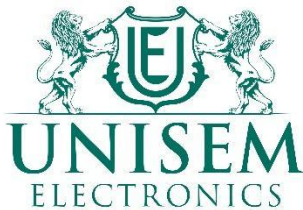


Fig11 : Recommended Reflow Profile

WIFI module installed note:

1. Please press 1 : 1 and then expand outward proportion to 0.7 mm, 0.12 mm thickness When open a stencil
2. Take and use the WIFI module, please insure the electrostatic protective measures.
3. Reflow soldering temperature should be according to the customer the main size of the products, such as the temperature set at $250 + 5^{\circ}\text{C}$ for the MID motherboard

For ordering USE_8189 please contact us on info@unisemindia.com.



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